

LAB EXPERIMENTS

Q19) Write a C program for monoalphabetic cipher is that both sender and receiver must commit the permuted cipher sequence to memory. A common technique for avoiding this is to use a keyword from Which the cipher sequence can be generated. For example, using the keyword CIPHER, write out the keyword followed by unused letters in normal order and match this against the plaintext letters:

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

cipher: C I P H E R A B D F G J K L M N O Q S T U V W X Y Z

Aim:

To write a C program to implement a **Keyword-based Monoalphabetic Substitution Cipher**, where the cipher alphabet is generated using the keyword **CIPHER** followed by the remaining unused letters of the alphabet.

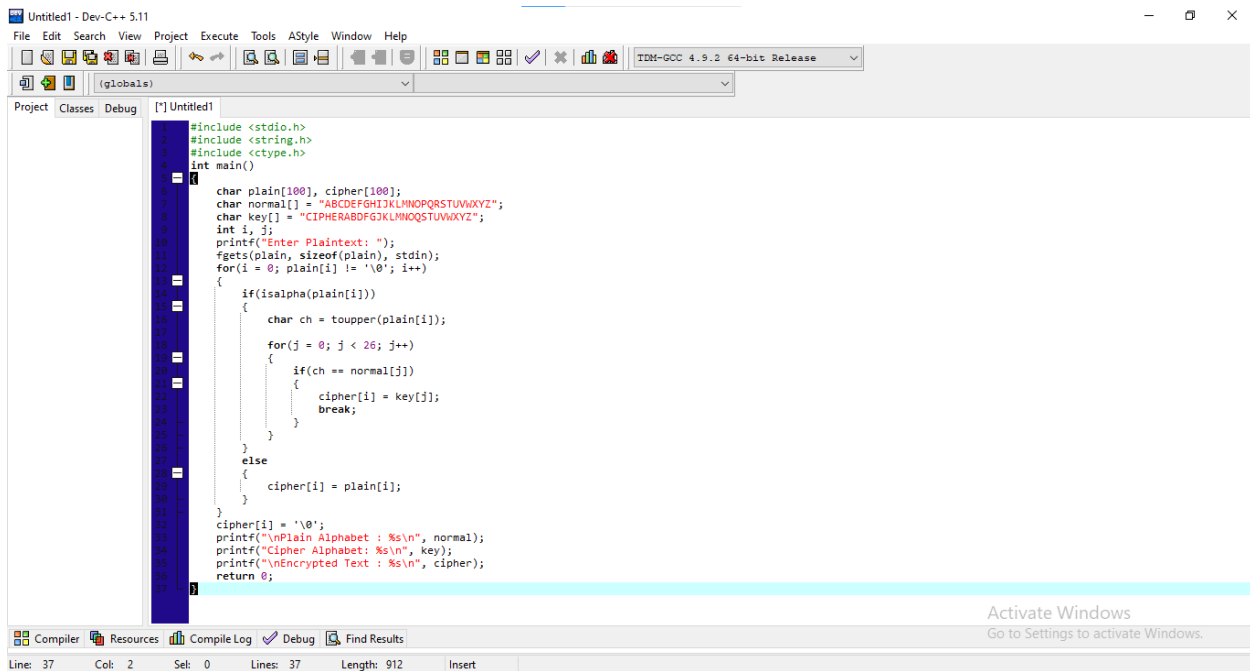
Algorithm:

1. Start the program.
2. Enter the plaintext message.
3. Define the plaintext alphabet as **A–Z**.
4. Generate the cipher alphabet using the keyword **CIPHER** followed by the remaining unused letters:
CIPHERABDFGJKLMNOQSTUVWXYZ
5. Read each character of the plaintext.
6. If the character is an alphabet, find its position in the plaintext alphabet.
7. Replace it with the corresponding character from the generated cipher alphabet.
8. Leave non-alphabetic characters unchanged.
9. Display the encrypted ciphertext.
10. Stop the program.

Result:

Thus, the C program to implement a **Keyword-based Monoalphabetic Substitution Cipher** using the keyword "**CIPHER**" was successfully executed, and the corresponding encrypted ciphertext was obtained.

Code:

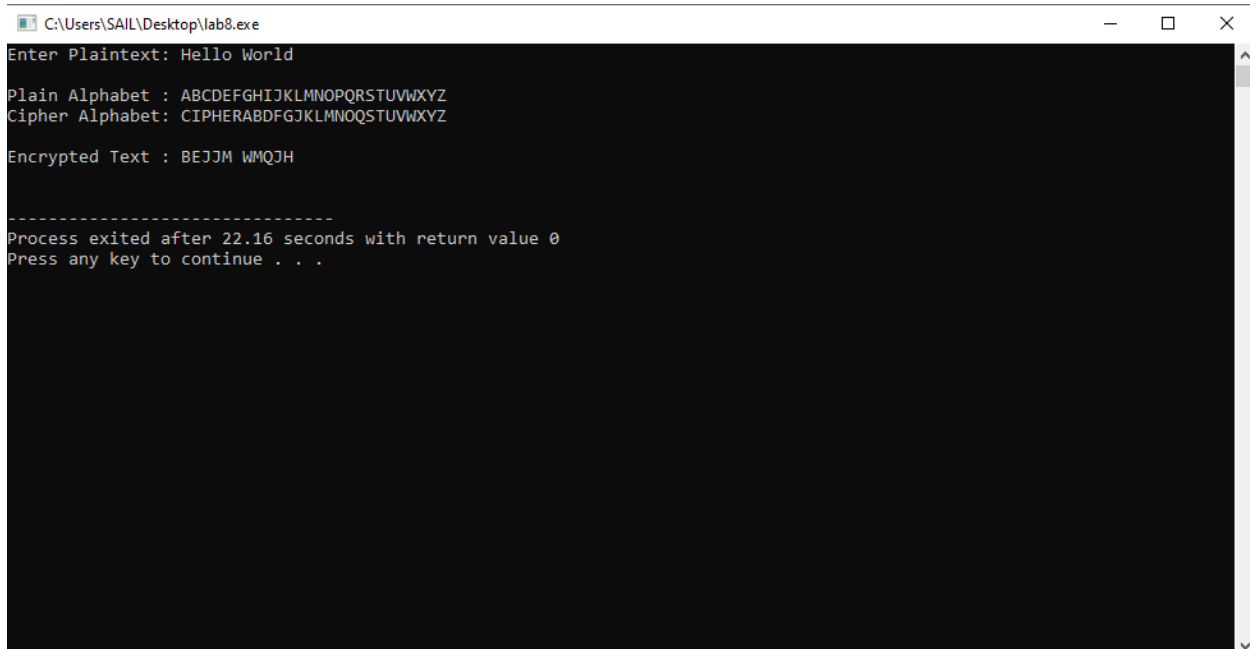


```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4 int main()
5 {
6     char plain[100], cipher[100];
7     char normal[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
8     char key[] = "CIPHERABDFGJKLMOQSTUVWXYZ";
9     int i, j;
10    printf("Enter Plaintext: ");
11    fgets(plain, sizeof(plain), stdin);
12    for(i = 0; plain[i] != '\0'; i++)
13    {
14        if(isalpha(plain[i]))
15        {
16            char ch = toupper(plain[i]);
17            for(j = 0; j < 26; j++)
18            {
19                if(ch == normal[j])
20                {
21                    cipher[i] = key[j];
22                    break;
23                }
24            }
25        }
26        else
27        {
28            cipher[i] = plain[i];
29        }
30    }
31    cipher[i] = '\0';
32    printf("\nPlain Alphabet : %s\n", normal);
33    printf("Cipher Alphabet: %s\n", key);
34    printf("\nEncrypted Text : %s\n", cipher);
35    return 0;
36 }
```

Activate Windows
Go to Settings to activate Windows.

Line: 37 Col: 2 Sel: 0 Lines: 37 Length: 912 Insert

Output:



```
C:\Users\SAIL\Desktop\lab8.exe
Enter Plaintext: Hello World

Plain Alphabet : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher Alphabet: CIPHERABDFGJKLMOQSTUVWXYZ

Encrypted Text : BEJJM WMQJH

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Process exited after 22.16 seconds with return value 0
Press any key to continue . . .
```